



Sir Syed University of Engineering & Technology, Karachi

CS-468

Software Quality Assurance & Testing

Week 2

Session 2

... 9 Causes of software errors ...

- **coding errors**
 - misunderstanding the design documentation
 - linguistic errors in the programming languages
 - errors in the application of development tools
 - errors in data selection

... 9 Causes of software errors ...

- **Non-compliance with documentation and coding instructions.**
 - Team members who need to coordinate their own codes with code modules developed by “non-complying” team members can be expected to encounter more than the usual number of difficulties when trying to understand the software developed by the other team members.
 - If the quality of the “non-complying” software is acceptable, future handling of this software (by the development and/or maintenance teams) is expected to increase the rate of errors:
 - Individuals replacing the “non-complying” team member (who has retired or been promoted) will find it difficult to fully understand his or her work. The design review team will find it more difficult to review a design prepared by a non-complying team.
 - The test team will find it more difficult to test the module; consequently, their efficiency is expected to be lower, leaving more errors undetected. Moreover, team members required to correct the detected errors can be expected to encounter greater difficulties when doing so. They may leave some errors only partially corrected, and even introduce new errors as a result of their incomplete grasp of the other team members’ work.
 - Maintenance teams required to contend with the “bugs” detected by users and to change or add to the existing software will face difficulties when trying to understand the software and its documentation. This is expected to result in an excessive number of errors and the expenditure of an excessive amount of maintenance effort.

... 9 Causes of software errors ...

- **Shortcoming of the testing process.**
 - Shortcomings of the testing process affect the error rate by leaving a greater number of errors undetected or uncorrected.
 - These shortcomings result from the following causes:
 - Incomplete test plans leave untreated portions of the software or the application functions and states of the system.
 - Failures to document and report detected errors and faults.
 - Failure to promptly correct detected software faults as a result of inappropriate indications of the reasons for the fault.
 - Incomplete correction of detected errors due to negligence or time pressures.

... 9 Causes of software errors ...

— **Procedure errors.**

- Procedures direct the user with respect to the activities required at each step of the process.
- They are of special importance in complex software systems where the processing is conducted in several steps, each of which may feed a variety of types of data and allow for examination of the intermediate results.

... 9 Causes of software errors

– **Documentation errors.**

- Errors in the design documents and in the documentation integrated into the body of the software.
- These errors can cause additional errors in further stages of development and during maintenance.
- Another type of documentation error, one that affects mainly the users, is an error in the user manuals and in the “help” displays incorporated in the software. Typical errors of this type are:
 - Omission of software functions.
 - Errors in the explanations and instructions given to users, resulting in “dead ends” or incorrect applications.
 - Listing of non-existing software functions, that is, functions planned in the early stages of development but later dropped, and functions that were active in previous versions of the software but cancelled in the current version.